

# Calculus III – Mock Final 1

Suggested time: 90 minutes    Total: 10 points

*No solutions are included in this booklet.*

**Question 1 – Derivatives and constrained extrema (2 points).** Let

$$f(x, y) = x^2y + e^{xy}.$$

- (a) Find the directional derivative of  $f$  at  $(1, 0)$  in the direction from  $(1, 0)$  toward  $(4, 4)$ .
- (b) Find the absolute maximum and absolute minimum of

$$g(x, y) = x^2 + y^2 - 2x - 4y$$

on the closed disk  $D = \{(x, y) : x^2 + y^2 \leq 9\}$ . Check interior critical points and use Lagrange multipliers on the boundary.

**Question 2 – Multiple integrals and polar/spherical coordinates (2 points).**

- (a) Evaluate

$$\iint_D (x^2 + y^2)e^{-(x^2+y^2)} dA,$$

where  $D$  is the upper half of the annulus  $1 \leq x^2 + y^2 \leq 4$ .

- (b) Using spherical coordinates, evaluate

$$\iiint_E (x^2 + y^2 + z^2) dV,$$

where  $E$  is the upper half-ball  $x^2 + y^2 + z^2 \leq 4$ ,  $z \geq 0$ .

**Question 3 – Scalar line integral and Green's theorem (3 points).**

- (a) Let  $C$  be the helix

$$\mathbf{r}(t) = (2 \cos t, 2 \sin t, t), \quad 0 \leq t \leq \pi.$$

Evaluate the scalar line integral

$$\int_C (x^2 + y^2 + z) ds.$$

- (b) Let  $\mathbf{F} = (P, Q) = (x - y, x + y)$  and let  $C$  be the positively oriented boundary of the region enclosed by  $y = x^2$  and  $y = 2x$ . Evaluate

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

using Green's theorem.

**Question 4 – Flux / surface integral of type 2 (3 points).** Let

$$\mathbf{F}(x, y, z) = (x, y, z^2).$$

Let  $S$  be the part of the cone  $z = \sqrt{x^2 + y^2}$  between  $z = 1$  and  $z = 3$ , oriented *downward*. Evaluate the flux

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS.$$

State in one sentence what the sign of your answer means geometrically.

**Calculus III – Mock Final 2****Suggested time: 90 minutes    Total: 10 points**

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**Question 1 – Derivatives and constrained extrema (2 points).**

(a) For

$$f(x, y, z) = xe^{yz} + y^2z,$$

find the directional derivative at  $(1, 0, 2)$  in the direction of the vector  $\langle 2, -1, 2 \rangle$ .

(b) Find the absolute maximum and minimum of

$$h(x, y) = xy$$

on the closed region  $D = \{(x, y) : x^2 + 4y^2 \leq 4\}$ .**Question 2 – Multiple integrals (2 points).**

(a) Use polar coordinates to evaluate

$$\iint_D \sqrt{x^2 + y^2} \, dA,$$

where  $D$  lies inside  $x^2 + y^2 = 4$  and above the line  $y = 1$ .

(b) Use spherical coordinates to find the volume of the solid inside the sphere

$$x^2 + y^2 + z^2 = 9$$

and above the cone

$$z = \sqrt{x^2 + y^2}.$$

**Question 3 – Scalar line integral and Green's theorem (3 points).**(a) Let  $C$  be the curve

$$\mathbf{r}(t) = (t, t^2, t^3), \quad 0 \leq t \leq 1.$$

Evaluate

$$\int_C x \, ds.$$

(b) Let  $C$  be the positively oriented boundary of the rectangle with vertices  $(0, 0), (3, 0), (3, 2), (0, 2)$ .  
Use Green's theorem to evaluate

$$\oint_C (x^2 - y) \, dx + (x + y^2) \, dy.$$

**Question 4 – Flux and physical meaning (3 points).** Let

$$\mathbf{F} = (x^2, y^2, z).$$

Let  $S$  be the upward-oriented part of the paraboloid

$$z = 4 - x^2 - y^2, \quad z \geq 0.$$

Evaluate

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

Interpret the integral as the net rate at which a fluid with velocity field  $\mathbf{F}$  crosses  $S$ .

**Calculus III – Mock Final 3****Suggested time: 90 minutes    Total: 10 points**

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**Question 1 – Directional derivative and Lagrange multipliers (2 points).**

- (a) Find the maximum rate of increase of

$$f(x, y) = xe^{-y} + ye^{-x}$$

at  $(0, 0)$  and give the direction in which it occurs.

- (b) Find the absolute extrema of

$$g(x, y) = x + 2y$$

on the closed region

$$D = \{(x, y) : x^2 + y^2 \leq 4, y \geq 0\}.$$

Use Lagrange multipliers where appropriate and check all boundary pieces.

**Question 2 – Double/triple integrals (2 points).**

- (a) Evaluate

$$\iint_D \cos(x^2 + y^2) dA,$$

where  $D$  is the quarter disk  $x^2 + y^2 \leq \pi$ ,  $x \geq 0$ ,  $y \geq 0$ .

- (b) Using spherical coordinates, evaluate

$$\iiint_E z^2 dV,$$

where  $E$  is the ball  $x^2 + y^2 + z^2 \leq a^2$ .**Question 3 – Line integrals and Green's theorem (3 points).**

- (a) Let
- $C$
- be the quarter-circle
- $x^2 + y^2 = 4$
- in the first quadrant, traversed from
- $(2, 0)$
- to
- $(0, 2)$
- . Evaluate

$$\int_C (x + y) ds.$$

- (b) Let
- $C$
- be the positively oriented circle
- $x^2 + y^2 = 4$
- . Use Green's theorem to evaluate

$$\oint_C (x^3 - y) dx + (x + y^3) dy.$$

**Question 4 – Vector surface integral / flux (3 points).** Let

$$\mathbf{F} = (xz, yz, z).$$

Let  $S$  be the upper hemisphere

$$x^2 + y^2 + z^2 = 4, \quad z \geq 0,$$

oriented outward. Evaluate

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS.$$

You may parameterize the hemisphere directly or close the surface with the disk  $z = 0$  and use an appropriate theorem if it has been covered in your course.

**Calculus III – Mock Final 4****Suggested time: 90 minutes    Total: 10 points**

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**Question 1 – Derivatives and absolute extrema (2 points).**

(a) Let

$$f(x, y) = \ln(x^2 + y^2).$$

Find the directional derivative at  $(1, \sqrt{3})$  in the direction making angle  $\pi/3$  with the positive  $x$ -axis.

(b) Find the absolute maximum and minimum of

$$g(x, y) = x^2 - y^2$$

on the closed disk  $x^2 + y^2 \leq 4$ .**Question 2 – Polar and spherical coordinates (2 points).**

(a) Evaluate

$$\iint_D xy \, dA,$$

where  $D$  is the sector  $1 \leq x^2 + y^2 \leq 9$ ,  $0 \leq y \leq x$  in the first quadrant.(b) Find the volume inside the sphere  $x^2 + y^2 + z^2 = 16$  and between the cones

$$z = \sqrt{x^2 + y^2} \quad \text{and} \quad z = \sqrt{3} \sqrt{x^2 + y^2},$$

using spherical coordinates.

**Question 3 – Scalar line integral and Green's theorem (3 points).**

(a) Let

$$\mathbf{r}(t) = (3t, 4t, 5t), \quad 0 \leq t \leq 1.$$

Evaluate

$$\int_C (x + y + z) \, ds.$$

(b) Let  $C$  be the positively oriented boundary of the region bounded by  $y = 0$ ,  $y = x$ , and  $x = 2$ . Use Green's theorem to evaluate

$$\oint_C (2x - y^2) \, dx + (x^2 + 3y) \, dy.$$

**Question 4 – Flux through a graph (3 points).** Let

$$\mathbf{F} = (y, -x, z^2).$$

Let  $S$  be the portion of the plane

$$z = 4 - x - y$$

above the triangular region  $D = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 4\}$ , oriented upward. Evaluate

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

## Calculus III – Mock Final 5

Suggested time: 90 minutes    Total: 10 points

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### Question 1 – Directional derivative and constrained optimization (2 points).

- (a) The temperature is

$$T(x, y, z) = 100e^{-(x^2+2y^2+z^2)}.$$

At  $P = (1, 1, 0)$ , find the directional derivative in the direction from  $P$  toward  $Q = (3, 0, 2)$ .

- (b) Find the points on the ellipse

$$x^2 + 2y^2 = 6$$

at which

$$f(x, y) = x + y$$

has its absolute maximum and minimum.

### Question 2 – Multiple integrals and coordinate changes (2 points).

- (a) Evaluate

$$\iint_D \frac{1}{1+x^2+y^2} dA,$$

where  $D = \{(x, y) : 1 \leq x^2 + y^2 \leq 4\}$ .

- (b) Using spherical coordinates, evaluate

$$\iiint_E \sqrt{x^2 + y^2 + z^2} dV,$$

where  $E$  is the solid inside the sphere  $x^2 + y^2 + z^2 \leq 9$  and above the  $xy$ -plane.

### Question 3 – Line integrals and Green's theorem (3 points).

- (a) Let  $C$  be the curve

$$\mathbf{r}(t) = (t, \cos t, \sin t), \quad 0 \leq t \leq \frac{\pi}{2}.$$

Evaluate

$$\int_C yz \, ds.$$

- (b) Let  $\mathbf{F} = (x + y^2, x^2 - y)$  and let  $C$  be the positively oriented boundary of the region enclosed by  $y = x^2$  and  $y = x$ . Evaluate

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

using Green's theorem.

### Question 4 – Flux, application, and geometric meaning (3 points). A fluid has velocity field

$$\mathbf{v}(x, y, z) = (x, y, 2z).$$

Let  $S$  be the outward-oriented lateral surface of the cylinder

$$x^2 + y^2 = 4, \quad 0 \leq z \leq 3$$

(without the top and bottom disks).

- (a) Compute the flux  $\iint_S \mathbf{v} \cdot \mathbf{n} \, dS$ .

- (b) Explain whether the result represents net flow into or out of the cylinder through its side.

## Design Notes and Topic Coverage

These mock exams were constructed to match the supplied MTH253 final-exam style: short computational questions, a 90-minute target, scalar line integrals, Green's theorem, and vector surface-flux problems. The supplied 2024–2025 exam had five 2-point questions; these practice versions consolidate the announced syllabus into four larger questions per test.

The exercise style is additionally inspired by the standard multivariable-calculus topics and graded problem-solving approach associated with James Stewart's *Calculus / Multivariable Calculus*, 8th edition. The questions above are newly written/adapted practice problems rather than verbatim reproductions of textbook exercises.

### Coverage in every test:

- Directional derivatives / gradient.
- Absolute extrema on closed bounded regions; Lagrange multipliers.
- Double or triple integrals with polar and/or spherical coordinates.
- Scalar line integrals ( $\int_C f \, ds$ ).
- Green's theorem.
- Vector surface integrals / flux, with geometric or physical interpretation.